

JATIN PAL

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Education

Bennett University(CGPA- 8)

Bachelor of Technology - Computer Science Engineering

August 2023 – June 2027

G.Noida, UP

Relevant Coursework

- Machine Learning
- Data Structures
- Deep Learning

Projects

Traffic Control using YOLO + Reinforcement Learning | Python, PyTorch, OpenAI Gym

- Designed an intelligent traffic management system integrating YOLOv8 object detection with reinforcement learning for adaptive signal control.
 - Built a simulation environment using SUMO + OpenAI Gym to train the RL agent on real-world traffic patterns.
 - Achieved significant reduction in vehicle waiting times by dynamically adjusting traffic signal timings based on live vehicle counts.
 - Demonstrated ability to combine real-time computer vision with decision-making AI, highlighting potential applications in smart cities and IoT-based traffic systems.
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Brain Tumor MRI Detection | Python, TensorFlow/Keras, OpenCV

- Developed a CNN-based medical image classification model to detect brain tumors from MRI scans.
 - Implemented preprocessing techniques such as normalization, resizing, and augmentation to improve model generalization.
 - Achieved high accuracy using transfer learning (ResNet, VGG) and validated performance with cross-validation.
 - Delivered a prototype system capable of assisting radiologists with faster and more reliable diagnostic support in healthcare applications.
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Stock Price Predictor | Python, Scikit-learn, LSTM, Pandas, Matplotlib

- Built a time-series forecasting model to predict stock price movements using historical market data.
- Engineered features such as moving averages, volatility indicators, and trading volume to enhance prediction accuracy.
- Trained LSTM and regression models, comparing performance across multiple evaluation metrics (MSE, RMSE, R^2).
- Designed a visualization dashboard to compare predicted vs. actual stock trends, showcasing practical use in financial decision-making.

Technical Skills

Languages: Python (NumPy, Pandas, Matplotlib, Seaborn, SciPy), SQL, Java, C++,javascript

Data Tools: Excel, Tableau, Jupyter Notebook, Google Colab, MySQL, MongoDB

Developer Tools: VS Code, Git,vercel,heroku,netlify,canva,figma

Libraries/Frameworks: Scikit-learn, TensorFlow, Keras, PyTorch, OpenCV, Streamlit,flask

Web Technologies: HTML5, CSS3, Streamlit Web App (Python-based UI rendering)

Certifications

- . **Programming in C++: A Hands-on Introduction Codio**
- . **Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization**
DeepLearning.AI Grade Achieved: 94.13%
- . **Operating Systems and You: Becoming a Power User Google Grade Achieved: 97.08%**
- . **Data Visualization with R IBM Grade Achieved: 91.33%**
- . **Peer-to-Peer Protocols and Local Area Networks University of Colorado System Grade Achieved: 93%**
- . **Project Initiation: Starting a Successful Project Google Grade Achieved: 92.20%**